# Learning DITA project

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# Chapter 1. What is DITA?

DITA, also known as Darwin Information Typing Architecture, is a standard that is based on the XML markup language. Developed in 2001 by IBM, DITA is a means of simplifying the process of creating and organizing technical information.

According to IBM, DITA is:

- **Darwin** - Information is organized in a hierarchy that is based on the idea of specialization and inheritance. This is derived from Charles Darwin's theory of evolutionary adaptation.
- **Information Typing** - Content is divided into reusable sections called topics.
- **Architecture** - The structure of DITA is expandable. New types of content can be created that aren't native to DITA. This means that DITA can be customized to suit your work style while still adhering to the same common tools and processes.

**Note:** For more information on DITA, visit IBM's DITA 101 Guide.

After the initial development of DITA, IBM donated it as an open source architecture and is now supported by the OASIS DITA Technical Committee.

## The Benefits of DITA

Using DITA is a versatile format that allows you to publish in different file types. For example, when working in programs such as Microsoft Word, Powerpoint, HTML, Adobe InDesign, Adobe FrameMaker, etc., your content is limited to their proprietary file types. DITA is able to take its content and publish them into HTML, PDF, Powerpoint, and mobile formats without having to tediously copy and paste content. This is because it is based on XML (eXtensible Markup Language).

- DITA is beneficial in terms of streamlining the process of writing and creating content.
- Standardizes the content, increasing the quality of the product.

## DITA and Markup Languages

A markup language is a way of tagging content, in plain text, that controls the layout and presentation of text and images in a digital document. The most common markup language is HTML.

DITA is based on XML, which is similar to HTML in that it uses angle brackets (`<` and `>`) for markup tags. An example of this is the `<p>` tag that is used for the paragraph element.

Example: `<p>` Here is some text. `</p>`

In both HTML and in XML, the tags can also have attributes, in the form **attribute="value"**, that provide further information about the tag.

Example: `note type="caution"> Failure to follow these steps can result in data loss.</note>`

The differences between XML and HTML:

| XML | HTML |
|---|---|
| Is strict when it comes to closing tags and quotation marks. | Is forgiving when you forget to close tags or quotes. |
| Uses predefined set of tags (`<p>`, `<body>`. `<span>`, etc.) | Tags are defined in separate file and maintained by an information architect. |

| XML | HTML |
| --- | --- |
| The name of the root tag depends on the type of topic. Example: `<concept>`, `<task>`, or `<reference>`. | The outer-most or “root” tag is `<html>`. |

DITA XML is important because every bit of content is stored within tags, such as titles, paragraphs, and lists, which allows you to classify your content depending on the topic type. These tags must be put in a certain way according to the rules established by DITA, in turn, standardizing your content.

The three major things that DITA XML does is:

- It organizes your content into topics.
- It separates content from format, making it portable and versatile. Your words and your format are no longer combined.
- It allows the content to be transformed into many other formats.

## Structured Authoring

Structured authoring is a workflow that allows the writer to define and apply consistent organization of information in documents. This methodology uses a topic-based structure, software-enforced rules, and allows for content reuse. The elements defined in DITA must adhere to a strict hierarchy and relationship. This means that each element has a specific set of elements that may contain it, and a certain set of elements that can be contained by it.

A `<table>` can have a head element that must be followed by one or more rows. The rules established by the software disallows the creation of a table with no rows or a table that contains a head that is not at the top of the table. Rules like these ensure consistency throughout the document.

The benefits of structured authoring:

- Collaboration
- Single-source output
- Consistency
- Easier content management
- Multiple format outputs

# Chapter 2. DITA Topics

Topics are the core components in DITA that allow for authoring and content reuse.

Topic-based authoring involves creating focused, independent pieces of content, each tackling a specific concept or question. A key benefit is the ability to use and reuse these topics in any order that you want. These topics are contained in DITA maps and allow you to organize the topics and specify the hierarchy as you see fit. For more information see:

In topic-based authoring, you must take the necessary steps to ensure that the content is reusable.

Reusable topics:

- Contain enough information to stand on its own.
- Target a single idea or answers a single question.
- Should not assume context and you should not assume that the reader has knowledge of what come before or after.
- A single file that contains a single topic.

All topics must contain a title element along with an id attribute. The topic id is usually generated automatically when a topic is created.

```
<topic id="sample_topic">

   <title>A title</title>

</topic>
```

## Generic Topic Structure

There are several types of topics, however, they are all built upon the generic topic type.

All topics follow this structure:

```
<topic id="sample_topic">

   <title>A title</title>

   <shortdesc>A short description that provides a 2-3 sentence summary of the topic.</shortdesc>

   <body>

        The bulk of the information in the topic.

   </body>

</topic>
```

### Concept Topics

Answers the question "Why?" and provides supportive information about a subject that is important to convey to the reader.

Concept topics commonly contain lists and paragraphs of text, as well as notes, tables, and images that support the text content.

Commonly used elements used in a concept topic are:

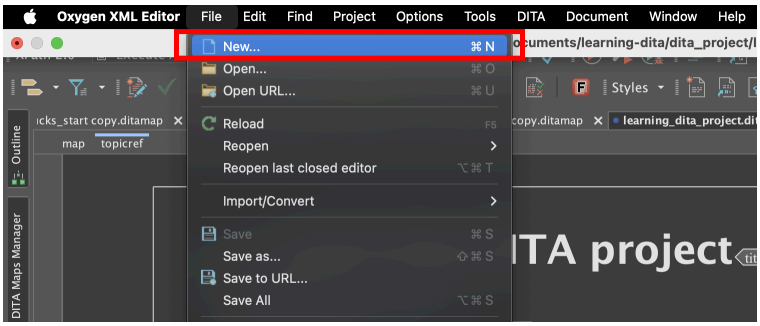
- <conbody> (the body of the concept topic)
- <p> (a paragraph)
- <ul> (an unordered or bulleted list)
- <ol> (an ordered or numbered list)

- <li> (a list item inside a <ul> or <ol>)
- <fig> (a figure, including an optional title)
- <image> (a graphic inside a figure, or inline in text)
- <section> (a subdivision in the topic, with an optional title)

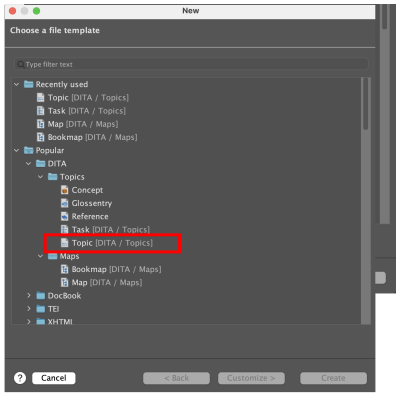
## How to Create a Topic in Oxygen XML Editor

Topics are integral to creating a document and will make up a majority of your content.

1. Navigate to **File > New**. A menu appears.




2. Select the **Topic** file template.




3. Add new content. A new blank file appears and if you are in **Author** mode, it will look like this:

# Notes, cautions, and warnings

The `<note>` element lets you create notes, cautions, warnings, and other guidance.

```
<note>
When the brownies are finished cooking, the pan will be very hot. Wear appropriate handwear when taking out
 the pan.

</note>
```

DITA defines a number of values for the type attribute:

- note (the default)
- attention
- danger
- caution
- important
- tip

```
<note type="warning">

Do not touch a hot pan without wearing any gloves.

We suggest wearing oven mitts or using a pot holder to handle the hot pan.

</note>
```

## Bulleted and numbered lists

The most common lists used in DITA are unordered (“bulleted”) lists (`<ul>`) and ordered (“numbered”) lists (`<ol>`).

Both ordered and unordered lists contain one or more list item (`<li>`) element.

An ordered list is useful for showing things that have a sequence.

```
<p>How to make brownies:</p>

<ol>

   <li>Line your baking pan with parchment paper. </li>

   <li>stir together the melted butter, granulated sugar, brown sugar, eggs, and vanilla extract until smooth
 and combined</li>

   <li>Fill baking pan 3/4 full.</li>

   <li>Unsweetened cocoa powder</li>

   <li>Bake for 20 minutes</li>

</ol>
```

An unordered list is useful for showing things that do not require a sequence.

```
<p>Brownie ingredients:</p>

<ul>

   <li>Egg</li>

   <li>Water</li>

   <li>Powdered sugar</li>

   <li>Unsweetened cocoa powder</li>

   <li>Oil</li>

   <li>Vanilla Extract</li>

</ul>
```

In addition to the unordered list (`<ul>`), DITA defines a simple list (`<sl>`) element, which you can use for lists that contain just a few words in each item.

```
<p>There are several types of brownies:</p>
<sl>
   <sli>Fudge brownies</sli>
   <sli>Oreo brownies</sli>
   <sli>Sprinkle brownies</sli>
   <sli>Cookies and cream brownies</sli>
</sl>
```

## Reusable Topics

One of the important benefits of using DITA is the reusability of topics. When creating topics it is important to remember:

- Topics should address a singe idea or answer a single question.
- They should be able to stand on its own
- They should not assume context of what comes before or after

**Note:** See Chapter 3: Reusing Content *(on page 10)* for more information on reusable topics.

## Map files

Map files are integral for creating documents using Oxygen and it is where all of your topics are stored and organized.

Map files are how you organize content for your output. It is similar to a table of contents, where it shows the hierarchy and relationship between topics.

Map files are made up mainly of the following components:

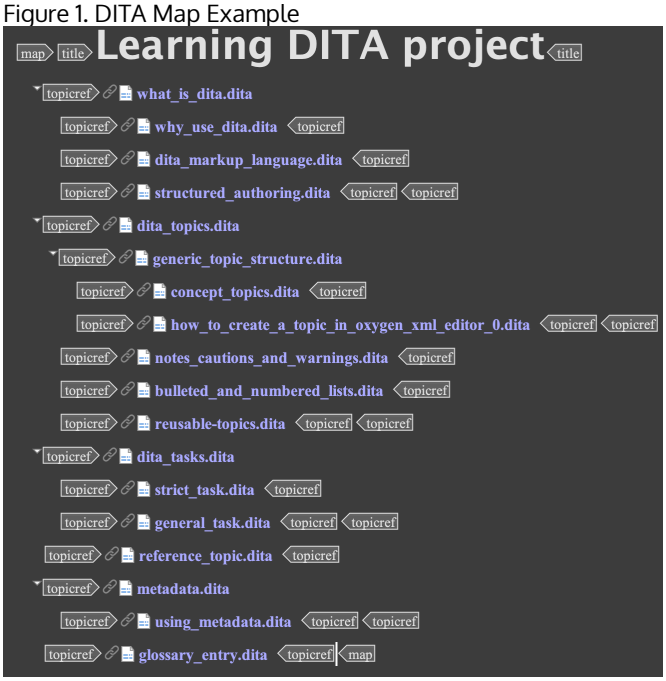
- topicref elements, which provide a link to a specific topic
- mapref elements, which provide a link to another map

In a map file, you put topicrefs in order from top to bottom to indicate sequence. To indicate hierarchy, you nest the topics. Here is an example of a map file in the **Text** view.

```
<map>
 <title>My map</title>
 <topicref href="brownies.dita">
   <topicref href="types_of_brownies.dita"/>
   <topicref href="ingredients.dita"/>
   <topicref href="pans.dita"/>
  </topicref>
</map>
```

Here is an example of a map file in the **Author** view in Oxygen XML:

Figure 1. DITA Map Example



# Chapter 3. Reusing Content

Reusing content, storing information in a single source and using it in other places, is one of the important benefits of using DITA.

Reusing content offers the several advantages:

- **Reduces replication**. Content reuse streamlines your workflow by eliminating the need to manually duplicate information or rewrite similar content in multiple places. This leads to increased efficiency and reduces the risk of errors.
- **Reduces cost**. Content reuse cuts down on the time wasted manually duplicating information, streamlining the creation process and eliminating the need for multiple reviews. Additionally, it significantly reduces localization costs. Replicated content requires individual translations for each instance, whereas a single, reusable source only needs translation once.
- **Allows for product or customer specific output**. Content reuse allows you to mix and match reusable topic modules across different documentation sets for various products or customer segments. Imagine a series of product manuals with some shared information and specific details unique to each product. Instead of creating separate manuals, you can combine common topics into a single map and then reference that map within each product-specific manual. This ensures you only write common information once, streamlining the content creation process.
- **Boosts consistency**. Manual content duplication invites human error. Multiple copies of the same information create a high risk of inconsistent updates, where some versions remain outdated while others are modified. This can lead to subtle discrepancies between copies over time, causing confusion and compromising accuracy. Maintaining a single source for reused content ensures consistent and up-to-date information wherever it appears.

## Determining Reusable Content

Content often contains reusable information, whether it's entire topics or individual elements like paragraphs or lists. Determining which content can help identify these reusable components, streamlining future content creation.

When analyzing content, consider the following:

- Sections of content that are exactly the same should be reused.
- Sections of content that do no match should not be reused.
- Sections of partially matching content can be reused after modifications. Here are examples of impartial matches:
    - Reusable Content with Naming Variations: Content is highly reusable, with the only exceptions being specific references to products, devices, model numbers, and companies.
    - Content Variations: While the core content remains the same, specific details like installation instructions or safety information may differ based on product versions.
    - Location: Content may vary due to different units of measurement that are intended for another country.
    - Order: Identical content may appear in a different order within different deliverables.
    - Specific details: The content may be similar, but vary in details such as the size or number of screws in a guide for building a table. This can be reused by removing specific details.

# Chapter 4. DITA Task Topics

Answers the question "How Do I?" using step-by-step instructions to guide the reader into completing a task.

DITA also allows step results, graphics, notes, and one level of substeps in task topics.

There are two types of task topics:

- Strict task
- General task

## Strict Task

Strict tasks are useful for procedural information, such as delivering step-by-step instructions or describing the actions that are necessary for completing something.

Strict tasks must:

- Appear in a specific order
- Contain only one `<example>` element
- Have two formal elements for steps - `<steps>` or `<steps-unordered>.`

Common strict task topic elements:

- `<taskbody>`
- `<steps>`
- `<cmd>`
- `<info>`
- `<stepresult>`
- `<stempxmp>`
- `<example>`

## General Task

The general task, also known as a "loose task" serves a similar purpose as the strict task as it instructs the user in completing a task. The difference being that this type of task is not limited in the number of elements that can be used. General tasks are typically used for process overviews.

Benefits of using a general task:

- More flexibility in order
- Allows for multiple `<example>` elements
- Allows for the use of `<step-informal>` element, which contains more varied content.

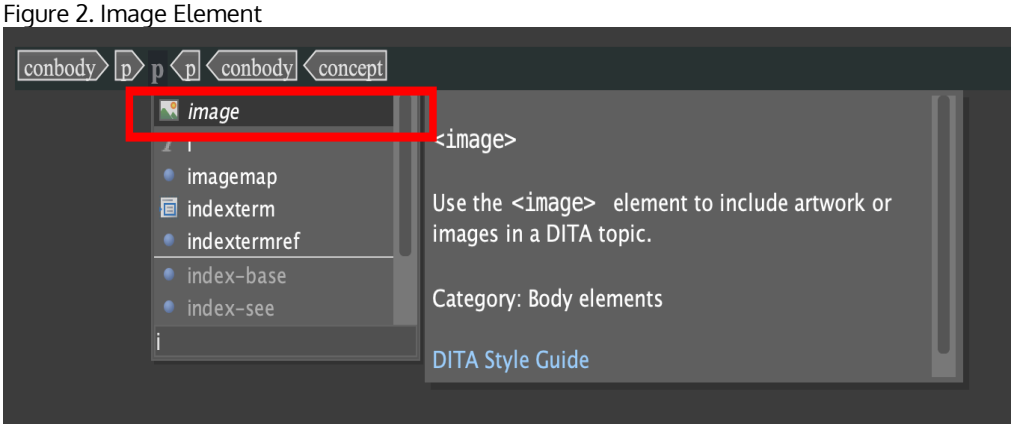
# Chapter 5. Adding an Image to a Concept Topic

Images can be added to a topic using the `<image>` element and it can be placed anywhere inside the `<conbody>` element.

This module will show you how to add an image to your concept topic using the **Author** view in Oxygen.

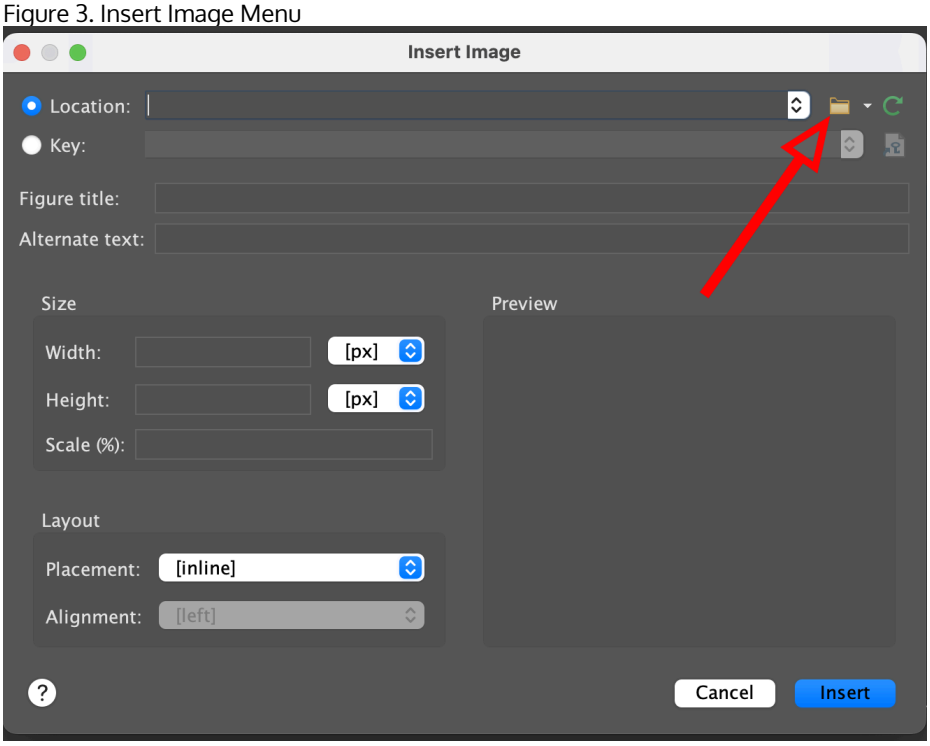
1. Press **Enter** while inside the `<<conbody>` element. A menu will appear with the available element options.
2. Type **image** and the element appears in the menu.

Figure 2. Image Element




3. Press **Enter** and the **Insert Image** window appears.
4. Click on the folder icon to browse for local files.

Figure 3. Insert Image Menu




   a. Add a **Figure Title** or **Alternative** if applicable.
5. Click **Insert** to add image.

# Chapter 6. The Table Element

The `<table>` element is used to organize relationships of information within columns and rows. DITA tables build upon the OASIS Exchange Table Model, adding DITA-specific features like accessibility, specialization, and conref support.

There are two elements that you can use to create tables in concept topics:

- `<simpletable>`
- `<table>`

## Simple Tables

You should use the <simpletable> element whenever you can, especially for short tables that are regular in structure and do not require a caption

<simpletable> elements:

| | |
|---|---|
| `<sthead>` | The container element for the header row of a `<simpletable>` element. The `<sthead>` element contains the `<stentry>` element and is optional. There can only be one `<sthead>` element in a `<simpletable>` element. |
| `<strow>` | The container element for the body rows of a `<simpletable>` element. The `<strow>` element contains the `<stentry>` element. A `<simpletable>` element can have one or more `<strow>` elements. |
| `<stentry>` | The container element for a single cell in a `<simpletable>` element. The `<stentry>` element contains the text of a cell in the `<simpletable>` element, which should be wrapped in a `<p>` element according to best practice. |

## Table Element

Use the `<table>` element for more complex tables or larger tables in your document. The `<table>` element allows your to add a title or caption to your table.

The elements contained in the <table> element include:

| | |
|---|---|
| `<title>` | The container element for the text of a `<table>` element's title or caption. A `<table>` element can only contain one `<title>` element. |
| `<tgroup>` | The container element for the main content of a `<table>` element. The `<tgroup>` element contains the `<colspec>`, `<thead>`, and `<tbody>` elements. A `<table>` element can contain one or more `<tgroup>` elements. |
| `<colspec>` | The element that provides information about the columns in a `<table>` element. The column specifications are defined using the colname (the name of the column), column (the order in which the column appears), and colwidth (the relative width of the column) attributes. |

| | |
|---|---|
| `<thead>` | The container element for the header in a `<table>` element. The `<thead>` element can contain one or more `<row>` elements and is optional. |
| `<tbody>` | The container element for the body rows in a `<table>` element. The `<tbody>` element can contain one or more `<row>` elements. |
| `<row>` | The container element for a row of cells in a `<table>` element. The `<row>` element contains one or more `<entry>` elements. |
| `<entry>` | The container element for a single cell in a `<table>` element. The `<entry>` element contains the text of a cell in the `<table>` element, which should be wrapped in a `<p>` element according to best practice. |

# Chapter 7. Cross-references

Cross-references allow you to create links from text to another location of the document. This includes linking to text, images, tables, or to external web pages.

All cross-references use the `<xref>` element. The href attribute specifies the target of the cross-reference. An example of this is this link that takes you to the "What is DITA" *(on page 3)* chapter.

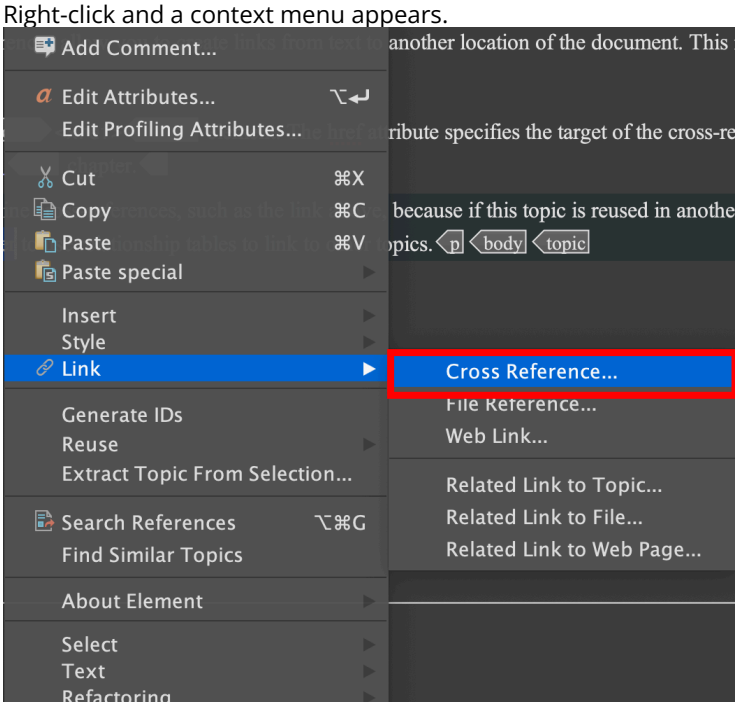
Generally it is bad practice creating inline cross-references, such as the link above, because if this topic is reused in another DITA map without the target link, the cross-reference will lead to a dead link. It is better to use relationship tables to link to other topics.

**Note:** For information on relationship tables visit the Relationship Tables page on the Oxygen XML Editor website.

## How to Create an Inline Cross-reference in Author View

This task teaches you how to create inline cross-references in your DITA topic.

1. Highlight the text you want to cross-reference.
2. Right-click and a context menu appears.




3. Click the folder icon and locate the topic you want to reference. The text hyperlinks and an `<xref>` tag surrounds the text if you have **Full Tags** enabled.

## External Cross-reference

This task teaches you how to create an external cross-references in your DITA topic in **Text View**.

1. Change your view to Text mode by selecting the **Text** button at the bottom of the editor.
2. Type in the `<xref> </xref>` tag.
3. Type `href="[your website link]" scope="external" format="html"` in the opening tag. A completed `xref` should look like this:

```
<xref href="http://www.youtube.com" scope="external" format="html"/>
```

# Chapter 8. Reference Topic

The purpose of a reference topic is to provide the user with the information and data to support them as they perform a task.

Reference topics answer the question "what is it?". Examples:

- What value do I enter here?
- What parameters should I use?
- What does this error indicate?
- What does this warning mean?

Reference topics are typically used to provide the reader with information as they need it when they are completing a task.

Two categories of reference topics:

- Single, stand-alone topics that contain a small number of tables or figures. Examples:
  - Matrix tables of devices and specific values or tolerances
  - Illustrations of device ports or pinouts
- Multiple, related topics that provide information about related items, one topic per item. Examples:
  - Command line reference
  - API function reference
  - Error code reference
  - Device data sheet

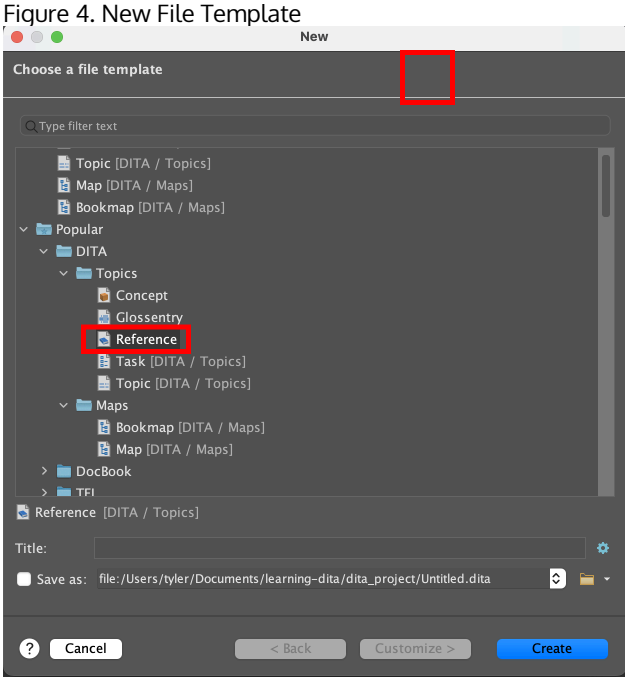
## How to Create a Reference Topic

This topic will provide you with instruction to create a reference topic using Oxygen.

At a minimum, the reference topic must contain a `<reference>` root element (with an id attribute) that contains a <title> element.

1. Click on the **New** button or go to **File > New**.
   A menu appears with various topic templates.
2. Select the **Reference** topic.

Figure 4. New File Template




**Tip:**

Following the <title> element and an optional <shortdesc> element, a <refbody> element contains the reference content. By design, the content in the <refbody> element is limited to a handful of elements. A reference topic is not as flexible in its content as a concept topic.

**Remember:**

The body of the reference topic can contain elements that have been introduced in previous courses: <table>, <fig> (figure) or <image>, <example>, and <section>.

# Chapter 9. Metadata

Metadata is information that describes other information. It contains details about the content, which makes it easier to understand. An example of metadata is the document properties found in Microsoft Word files, which provides the name of the creator and the date of the last modification.

Uses of metadata:

- Label information with a document type, such as “white paper” or “proposal”
- Associate information with a specific product number
- Indicate a user level, such as “beginner,” “intermediate,” or “expert”
- Indicate the audience for the content, such as “manager” or “employee”
- Provide status information for a topic, such as “draft,” “review,” or “final”
- Provide an expiration date when information is no longer valid

Metadata helps the user to:

- Find information efficiently
- Support governance effort (control when information is made public or removed from a web site)
- Personalize information (your reader could request only beginner information)
- Filter information for different delivery variants (such as Windows or Mac)
- Reuse information efficiently
- Indicate the language used in content
- Manage project status

DITA requires one piece of metadata. Every topic must have an id attribute. These IDs are often assigned automatically by authoring tools or content management systems (CMS).

```
<topic id="sample_topic">

   <title>A title</title>

   <body>

      ....

   </body>

</topic>
```

## Using Metadata

Metadata can be assigned to DITA content in a few ways:

- At the topic level
- At the element level
- At the map file level

At the topic level, DITA provides a <prolog> element in which you can store metadata for the entire topic.

```
<topic id="sample_topic">

  <title>Metadata example</title>

  <prolog>

    <author>Tyler Nguyen, English 382</author>

    <critdates>

      <created date="2024-05-19"/>
```

```
    </critdates>
  </prolog>
  <body>
    <p>Body content</p>
  </body>
</topic>
```

Here the author is specified in the `<author>` element, and `<critdates>` provides a spot for `<created>` and `<revised>` date elements.

Useful prolog elements:

- `<author>` (the content author)
- `<critdates>` (critical dates, such as `<created>` and `<revised>`)
- `<copyright>` (copyright year `<copyryear>`, and copyright holder `<copyrholder>`)
- `<vrm>` (product version, release, and modification information)

At the element level, you usually use attributes in elements to specify metadata.

```
<step>
    <cmd>Locate the brownie mix box.</cmd>
    <info audience="novice">Consult the side of the brownie mix box to determine
          how much milk and butter you need.</info>
</step>
<step>
    <cmd>Measure out the wet ingredients for your brownies. </cmd>
</step>
<step>
    <cmd> Pour in the mixing bowl.</cmd>
</step>
<step>
    <cmd>Put in baking sheet.</cmd>
</step>
```

In this example for making brownies, only novice bakers need to consult the box for accurate measurements. When generating the output, you can hide the `<info>` element that uses the "novice" metadata for an expert-level user.

At the map level, you can specify metadata in the `<topicref>` elements that reference topics. This allows you to suppress entire topics when you generate output.

```
<topicref href="abc.dita">
<topicref href="def.dita" audience="novice">
```

By default, DITA provides you with three attributes that support filtering or conditional processing. These attributes are:

- audience
- product
- platform

If you need additional or different filtering attributes, your information architect will need to specialize to define additional metadata. Common examples are:

- customer, for customer-specific information
- region, for information that applies only to specific geographic areas
- product-family, for information that applies to a group of products

# Chapter 10. Glossary Entry

Glossary entry topics are used to define specific terms or concepts within your content.

Each glossary entry is a separate file that contains a single term and definition.

It is possible to maintain multiple glossary entries in a single file, a glossary group, using the glossary group topic type. This topic type can contain one or more glossary entry.

Reasons to have stand-alone glossary terms:

- Reuse - Once a glossary entry is created, the same entry can be used in other product manuals.
- Ownership - Glossary entries within a glossary group are specific to certain product or to the team that created the group. It can be difficult to merge new entries into an existing glossary group.
- Localization - Maintaining entries that are separate files makes it easier to sort translated glossary terms.
- References - DITA offers multiple elements for linking text to glossary definitions. References are easy to build when files are separate compared when to glossary groups.

# Chapter 11. Publishing DITA Files

If you are creating content using DITA, chances are you are doing it with the intention of generating an output in the form of a PDF or HTML. There are numerous publishing engines, but for the sake of this guide, we will be using the DITA Open Toolkit and Oxygen XML to generate our output.

When choosing a publishing environment, it is important to consider:

- **How often content is updated**. Documents that updated frequently will require you to generate output more often. In this case, it is best to use enterprise-level publishing software.
- **How much content there is** . While open-source tools might suffice for a couple of small manuals, enterprise-level environments offer the necessary capabilities and scalability to handle hundreds of manuals with thousands of pages, ensuring efficient and streamlined content management.
- **How many outputs are required**. For a small number of outputs, a standalone environment may be adequate. However, publishing a large volume necessitates a commercial system with the capacity and tools to handle extensive output creation.

## Publishing with the DITA Open Toolkit

The DITA Open Toolkit or DITA OT is an open-source publishing engine for DITA content that is maintained by a group of volunteers. DITA OT uses plugins to generate DITA into PDF, HTML, WebHelp, and other outputs. Plugins can be customized to change the appearance of the output. Best of all, DITA OT is a free tool for anyone to use. You just need to install the latest version of Java and DITA OT, which can be found on the DITA OT website.

DITA OT can be used alongside other DITA authoring tools to create content or it can be used alone. The most common uses for DITA OT:

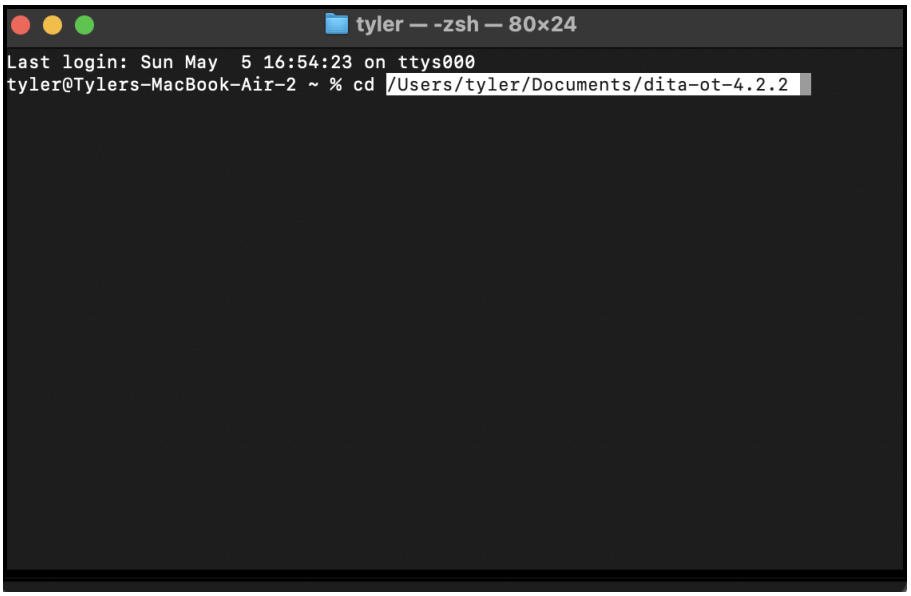
- **Small content volume.** If a company is generating only a few documents and are not regularly updated.
- **Experimental projects.** DITA OT can be useful when showcasing the benefits of using DITA to people in your company.
- **Small teams.** A small team of writers may not need the robust features of an enterprise-level authoring tool. It would be more cost effective to use an open-source publishing engine.
- **Students learning DITA.** Standalone DITA OT is the best solution for a student because it is free and lightweight.

**Note:** Using a standalone DITA OT can be less intuitive than using commercial software because you are required to use the Terminal or Command Line to generate output.

### How to Publish Using the DITA Open Toolkit

This guide will show you how to create a PDF output using the DITA OT.

1. Open the command line.
    a. On Mac: Launch the Terminal app.
    b. On Windows: Click **Start > Run** and type **cmd**.
2. Type `cd` followed by the filepath to your **dita-ot-2.4.4** folder.




**Note:** For a easier way to get your filepath in the command line, locate the file in your file explorer and drag the folder into the command line window.

3. Type this command:
   a. Mac: `./bin/dita -i [filepath]/test.pdf -f pdf2 -o out/file_output -v`
   b. Windows: `bin\dita -i [filepath]/test.pdf -f pdf2 -o out/file_output -v`
4. Press **ENTER** to run the DITA OT.

   When the build has finished, look in the **file_output** folder to see your generated output. You should see a file called **test.pdf**.

**Note:** You can change the name of the folder by substituting the **`file_output`** with a name of your choosing.

## DITA OT Command and Parameters

The standard commands and parameters of a basic DITA OT build are:

- For Mac: `./bin/dita -i [filepath]/test.pdf -f pdf2 -o out/file_output -v`
- For Windows: Windows: `bin\dita -i [filepath]/test.pdf -f pdf2 -o out/file_output -v`

The meaning of the commands and parameters:

| Command or Parameter | Function |
|---|---|
| `./bin/dita(Mac) bin\dita (windows)` | The command that generates output from DITA. |
| `-i` (the input parameter) | Indicates that the filepath that follows it to a DITA map or file needs to be transformed into output. |
| `-f` (the transtype parameter) | Indicates that the string that follows it is the output transformation type (also called the “transtype”). Common transtype values include **pdf2** for PDF output, **xhtml** for HTML output, and **webhelp** for WebHelp output. |

| Command or Parameter | Function |
|---|---|
| `-o` (the output parameter) | Indicates where to store the generated output. This parameter is optional; if you do not use it to specify an output directory, the output is placed in the DITA OT's default "out" folder. |
| `-v` (the verbose parameter) | Indicates verbose output, which displays a log of the DITA OT build in the command window. This parameter is optional, but it is useful to watch the DITA OT run in realtime to help troubleshoot errors. |

**Note:** For more information on parameters, check out the DITA OT Parameter Reference page.

# Publishing with the Oxygen XML Editor

Publishing content using an authoring tool, such as Oxygen XML Editor, is much more common due to its accessibility. Unlike using standalone DITA OT, Oxygen has a graphical user interface and generating content is as easy as clicking a single button.

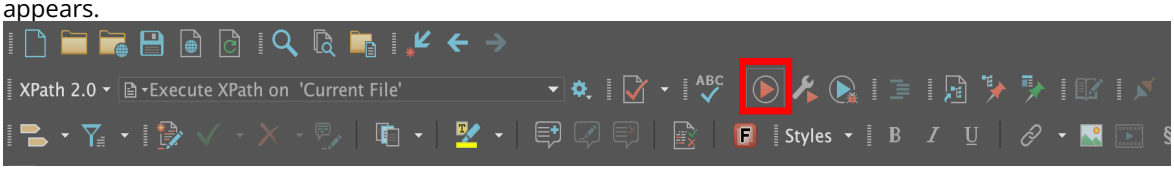
Benefits of using an authoring tool:

- **User friendly**. Using an authoring tool is much easier for someone who does not have experience using DITA or uses it infrequently.
- **Easy of use**. It is easier to customize and organize your output compared to standalone DITA OT.
- **Cost effective**. While Oxygen does have licensing costs, it can be a more budget-friendly option for smaller teams or those who don't require the comprehensive features of a component content management system (CCMS).

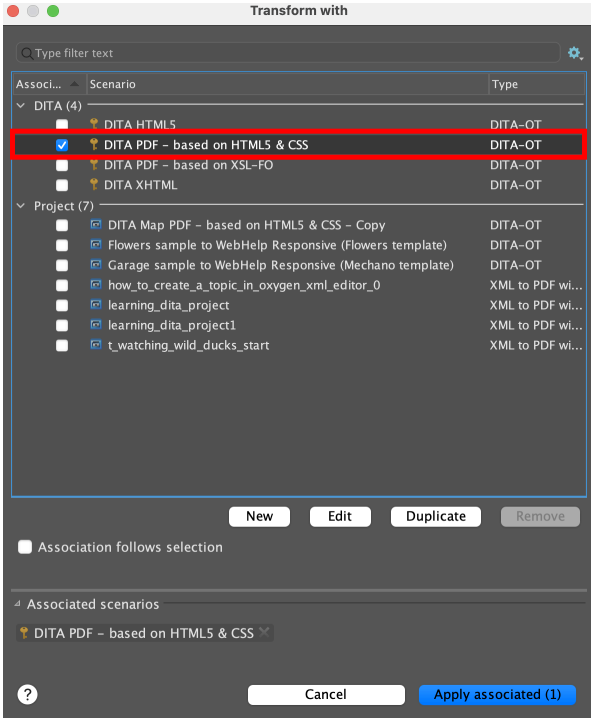
## How to Publish Using Oxygen EML Editor

This guide will show you how to create a PDF output using Oxygen XML Editor.

1. Open the DITA topic or map that you want to publish.
2. Click the **Apply Transformation Scenario(s)** button on the toolbar. The **Transform with** window appears.




3. Select the **DITA PDF - based on HTML5 & CSS** checkbox.



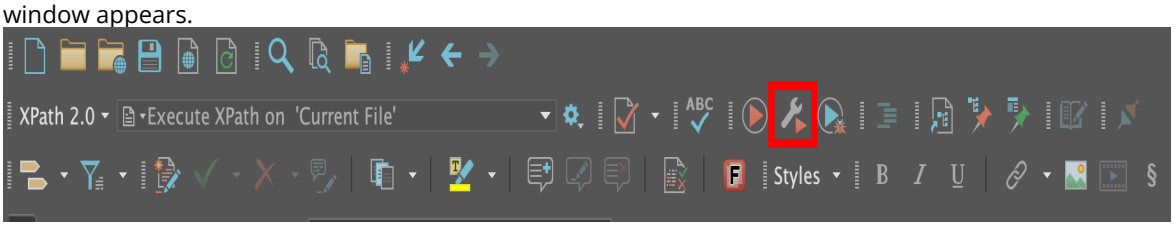

4. Click **Apply associated**. Your PDF will generate and open in the default PDF viewer on your system.

**Note:** It may take a few seconds for your PDF to be generated depending on the size of your project.

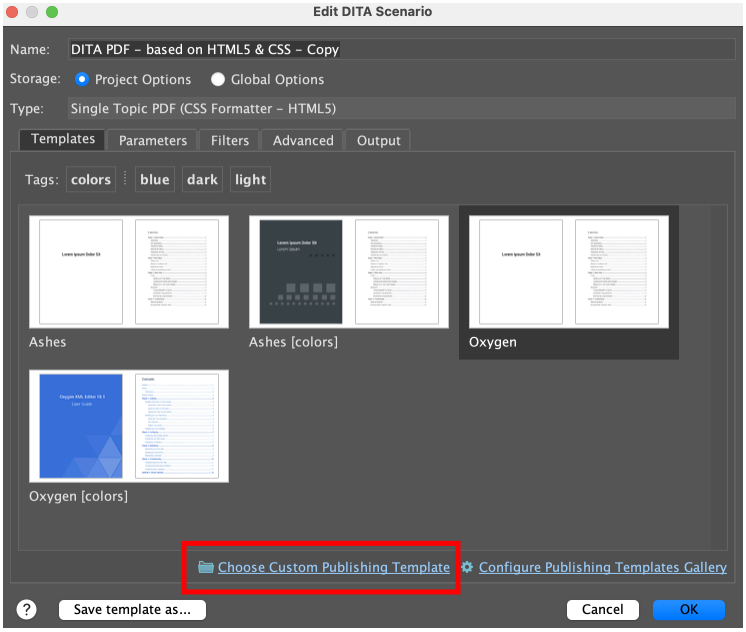
## How to Create Custom output

The default output for PDF is serviceable, but very plain. To make it more visually appealing, you will want to customize the appearance of the output. This guide will show you how to customize the output using **Styles Basket for Oxygen**.

1. Go to the Style Basket for Oxygen website.
2. Use the **Galleries and Styles** tab to customize your output.
3. Click **Download** once you are satisfied with your settings. Your style will download as a zip file.
4. Click on **Configure Transformation Scenario(s)** in Oxygen. The **Configure Transformation Scenario** window appears.




5. Select the **DITA PDF - based on HTML5 & CSS** checkbox and then click the **Duplicate** button. The **Edit DITA Scenario** window appears.
6. Click **Choose Custom Publishing Template** and locate the zip file that you downloaded. The style will appear as a template.




7. Publish your document to view your new style.

# Chapter 12. Appendix: Learning DITA Certificates

Here are the various certificates that I received from completing the LearningDITA course.

Figure 5. Introduction to DITA

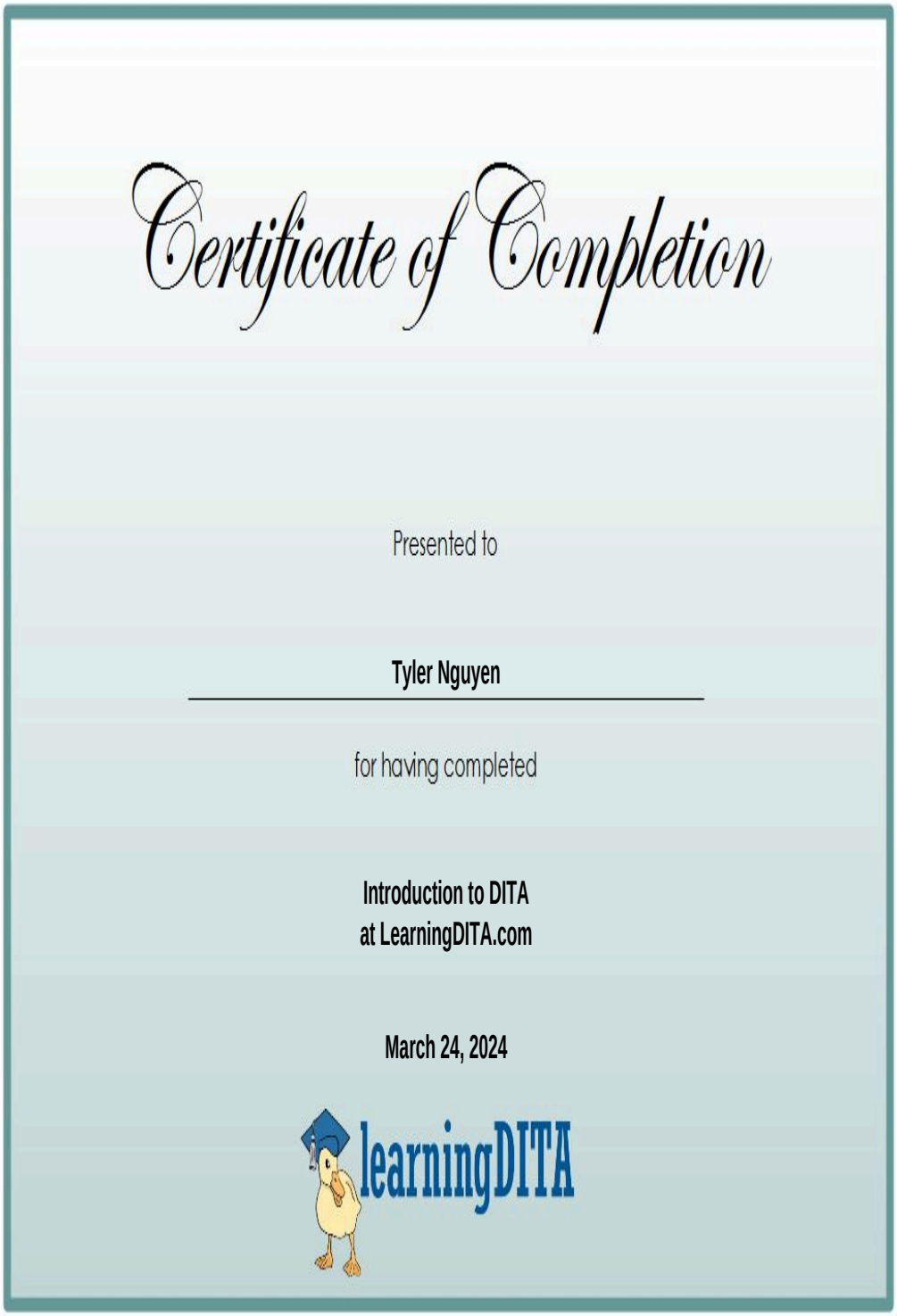


Figure 6. The DITA Concept Topic Type

Figure 7. The DITA Task Topic Type

Figure 8. The DITA Reference and Glossary Topic Type

Figure 9. Using DITA Maps and Bookmaps

Figure 10. Introduction to Reuse

Figure 11. Advance Reuse in DITA

Figure 12. Publishing Output from DITA Sources

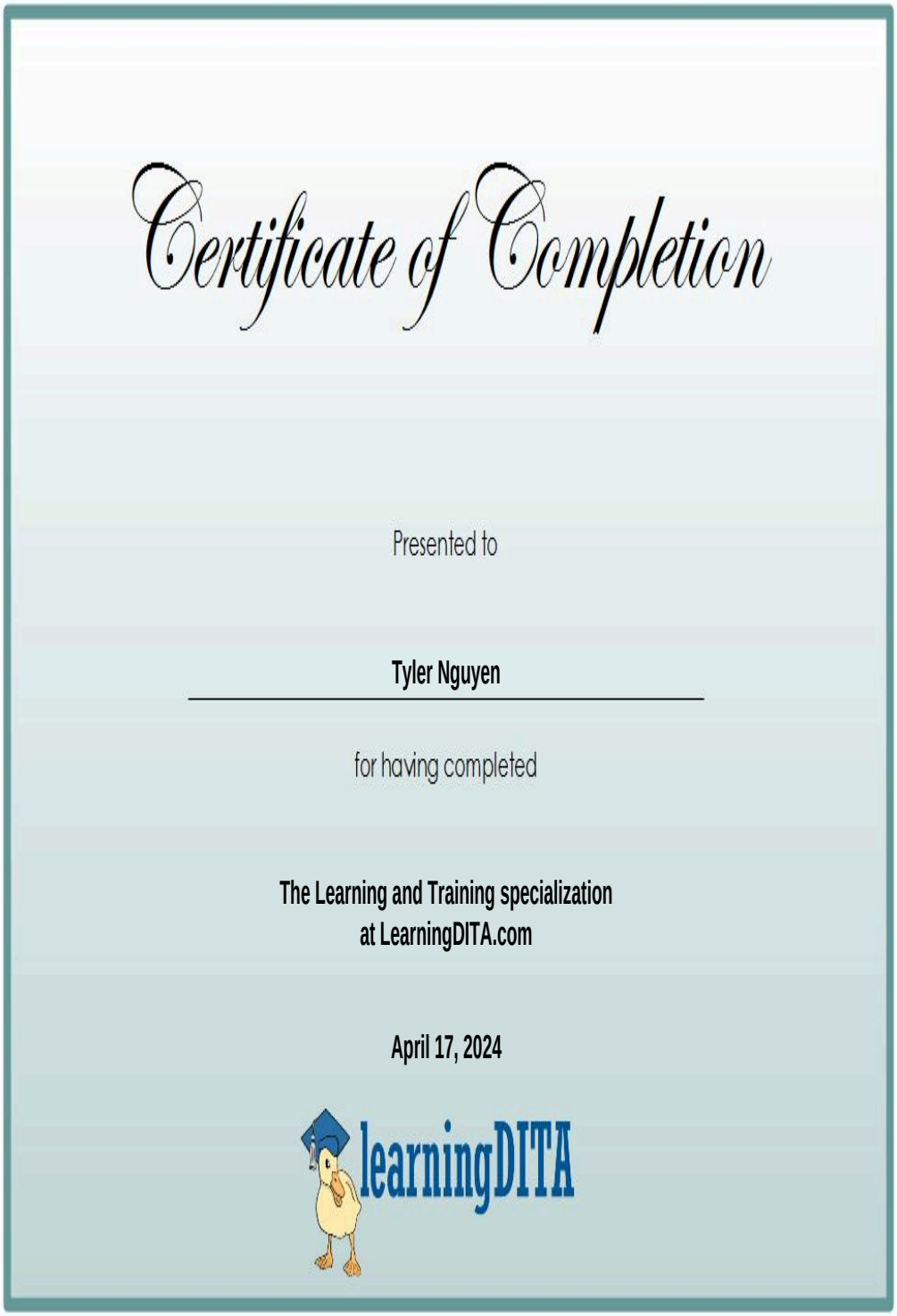



Figure 13. The Learning and Training Specialization